Wind Turbine Project: Part III FOWT Design at Bretagne Sud 1 Site

Louis Yue Ting Shing, *M2 STEEM, Ecole Polytechnique,* Luca Knab, *M2 STEEM, Ecole Polytechnique*

I. Introduction

A. Context and Motivation

FLOATING offshore wind turbine design depends on site-specific metocean conditions because wave forcing governs platform motions, mooring loads, accessibility, and survival, while wind forcing governs rotor thrust and the resulting global overturning demand on the support system. At Bretagne Sud 1, design decisions are therefore controlled by how representative sea states translate into local wave kinematics and coupled wind–wave loading at the turbine location.

The environmental inputs used in this report are taken directly from the ResourceCode hindcast at the Bretagne Sud 1 grid point analysed in Parts I and II, so the operational and 50-year storm sea states already correspond to the site vicinity rather than to a generic offshore climate.

Part I provided the statistical basis for selecting representative operational and extreme sea states, including return-level estimates and joint combinations of significant wave height and characteristic period used as design conditions. Part II used those sea states as inputs to ray-tracing simulations to contextualise how propagation effects and bathymetry can redistribute wave energy in the broader area, thereby motivating the use of directionally consistent and physically interpretable forcing cases for structural analyses.

The monopile calculations in Part III are retained as an analytical exercise rather than as a realistic foundation choice for $h = 93.32\,\mathrm{m}$. Fixed-bottom monopiles are generally associated with shallow sites, whereas floating concepts are typically considered at larger depths, so the monopile overturning-moment estimates are used here mainly to benchmark wave-induced loading levels and to separate wind- and wave-driven contributions in a controlled setting. The floating analyses are therefore the more representative feasibility component for Bretagne Sud 1, and they assess whether SPAR- and BARGE-type concepts can satisfy motion limits under the same site-derived sea states.

B. Statement of Purpose

This report addresses two connected design questions using the operational and 50-year storm sea states derived previously for Bretagne Sud 1. First, wave loads on an idealised fixed monopile of diameter $D = 10\,\mathrm{m}$ are estimated using Morison-type formulations, including checks of non-dimensional applicability and the computation of seabed shear force and overturning moment for representative regular-wave equivalents.

Second, surge, heave, and pitch motions of spar and barge floating offshore wind turbine concepts are analysed in the time domain using Python with wind and wave inputs consistent with the same sea states, and key response metrics are compared against the acceptance criteria provided in the practical material. Taken together, the analyses provide a quantitative link between the locally derived sea states and structural demand and operability, and they identify whether extreme-condition response is controlled by quasi-static loading or by resonance with platform natural periods.

C. Report Outline

Section II estimates wave-induced loads and seabed overturning moments on an idealised fixed monopile of diameter $D = 10\,\mathrm{m}$ using Morison-type formulations. The validity of the modelling assumptions is first assessed using non-dimensional parameters, after which hydrodynamic coefficients are selected, and the resulting wave-induced bending is compared with wind-induced overturning moments for the operational and 50-year storm sea states.

Section III analyses the dynamic response of floating offshore wind turbine concepts under the same environmental conditions. Time-domain simulations are used to compute surge, heave, and pitch motions for SPAR and BARGE configurations, extract key response metrics, and assess compliance with the motion acceptance criteria discussed in the practical material. Section IV concludes the report.

II. Analytical Estimation of Wave Loads on a Monopile

Wave loads on a monopile of diameter $10\,\mathrm{m}$ are evaluated using the Morison equation. Two representative sea states at the Bretagne Sud 1 site are considered: an operational condition and a 50-year extreme storm. The analysis first checks the Morison equation validity using non-dimensional parameters, then applies the selected hydrodynamic coefficients to compute inline force distributions, seabed overturning moments, and to compare wave- and wind-induced bending.

A. Wave and Structural Parameters

The academic DTU $10\,\mathrm{MW}$ wind turbine is located at the centre of the support structure, and its main inertial and aerodynamic properties are summarised in table I. These parameters define the gravity loads and aerodynamic thrust that act at the monopile head and therefore link the hydrodynamic loading to the global support-structure response.

Table I: DTU 10 MW turbine and tower properties used in the load calculations [1], [2]. Heights are given relative to mean sea level (MSL).

Quantity	Value
Rotor–nacelle assembly mass m_{RNA}	650 t
RNA centre of gravity z_{RNA}	120 m above MSL
Tower mass m_{tower}	600 t
Tower centre of gravity z_{tower}	50 m above MSL
Typical tower diameter	7 m
Maximum aerodynamic thrust $T_{\max}$	1500 kN at $z = 120$ m
Drag coefficient in parked conditions $C_{D,\text{turb}}$	0.01

The monopile foundation is represented by a vertical circular cylinder of external diameter $D = 10\,\text{m}$ installed at the Bretagne Sud 1 site, with the hub located at $z = 120\,\text{m}$ above mean sea level and the seabed at $z = -93.32\,\text{m}$. The overall geometrical arrangement together with the operational and extreme wind and wave conditions used in the load calculations is illustrated in figure 1.

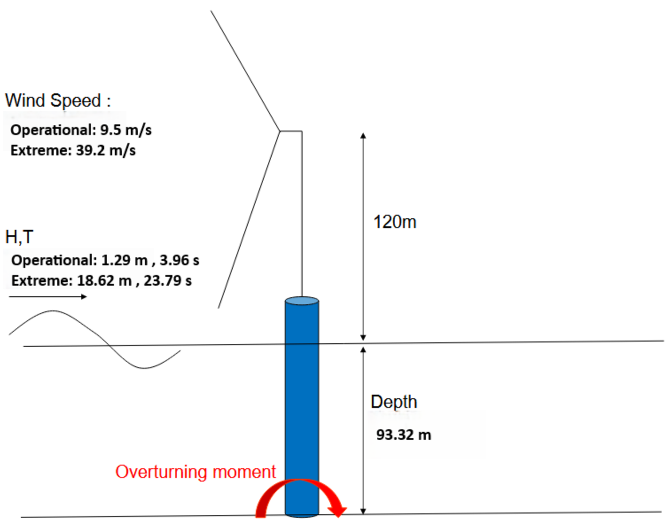

Figure 1: Schematic of the DTU 10 MW wind turbine on a 10 m diameter monopile at the Bretagne Sud 1 site, indicating water depth, hub height and the operational and extreme wind and wave conditions used in the Morison load calculations (adapted from [1] with metocean values updated for Bretagne Sud 1).

Two metocean conditions, consistent with the overturning-moment exercise, are given in table II. The still-water depth is $h = 93.32\,\text{m}$ at the turbine location. Each sea state is modelled as an equivalent regular wave with prescribed height H and period T and is associated with a concurrent hub-height wind speed U_{hub} that determines the aerodynamic thrust.

Table II: Representative local sea states and environmental parameters at the Bretagne Sud 1 monopile location. The same water depth h is used for both cases. Hub-height wind speeds U_{hub} at $z = 120\,\text{m}$ correspond to the mean operational and 50-year storm conditions at the site.

Case	h (m)	H (m)	T (s)	U_{hub} (m s^{-1})
Operational	93.32	1.29	3.96	9.49
Extreme storm	93.32	18.62	23.79	39.2

The wind and wave parameters in table II were derived from the 27-year ResourceCode hindcast at the Bretagne Sud 1 grid point described in Parts I and II of this project. For the operational case, the mean 10 m wind speed and the mean significant wave height and peak period were computed over the full record, giving a sea-level wind speed $U_{10,\text{mean}} = 7.22\,\text{m}\,\text{s}^{-1}$ and the wave pair $(H, T) = (1.29\,\text{m}, 3.96\,\text{s})$. For the extreme storm case, annual maxima of the 10 m wind speed were extracted and fitted with a block-maxima extreme value model to obtain the 50-year return wind speed at sea level, $U_{10,50} = 29.82\,\text{m}\,\text{s}^{-1}$, while a multivariate extreme value analysis of (H, T) yielded the 50-year storm pair $(H_{50}, T_{50}) = (18.62\,\text{m}, 23.79\,\text{s})$.
Sea-level winds were then extrapolated to the turbine hub height using the standard offshore power-law profile,

$$U(z) = U_{\text{ref}} \left(\frac{z}{z_{\text{ref}}} \right)^{\alpha}, \tag{1}$$

with reference height $z_{\text{ref}} = 10\,\text{m}$, shear exponent $\alpha = 0.11$ appropriate for neutral marine boundary layers, and hub height $z = 120\,\text{m}$. Applying this relation to $U_{10,\text{mean}}$ and $U_{10,50}$ gives the hub-height speeds $U_{\text{hub,op}} = 9.49\,\text{m}\,\text{s}^{-1}$ and $U_{\text{hub,ex}} = 39.2\,\text{m}\,\text{s}^{-1}$ reported in table II.

Fluid properties entering the Morison equation are summarised in table III. A constant seawater density and standard gravity are assumed for all calculations.

Table III: Constants used for the monopile load calculations.

Quantity	Value
Water density ρ	1025 kg m^{-3}
Gravitational acceleration g	9.81 m s^{-2}

B. Keulegan–Carpenter number

The suitability of the Morison formulation is assessed first using the Keulegan–Carpenter number KC, which compares the horizontal excursion of the water particles during one wave cycle with the characteristic size of the structure and thereby indicates the relative importance of inertia and drag in the inline force on a vertical cylinder.

The Keulegan–Carpenter number is defined as

$$KC = \frac{UT}{D},$$

where U is a characteristic horizontal flow velocity, T is the wave period, and D is the monopile diameter [3]. For wave-induced oscillatory flow, U is taken as the amplitude of the horizontal orbital velocity at still water level $z = 0$ from linear (Airy) wave theory. For a regular wave of height H, period T, wavenumber k, and water depth h, the horizontal velocity at elevation z is

$$u(z,t) = \frac{\pi H}{T} \frac{\cosh\big(k(z+h)\big)}{\sinh(kh)} \cos(kx - \omega t),$$

with angular frequency $\omega = 2\pi/T$ [3]. The velocity amplitude is obtained by setting the phase factor to its maximum value, $\cos(kx - \omega t) = 1$, because only the peak magnitude of the oscillatory motion is required for KC. This gives

$$U_m(z) = |u(z,t)|_{\max} = \frac{\pi H}{T} \frac{\cosh\big(k(z+h)\big)}{\sinh(kh)}.$$

Evaluating at the still water level $z = 0$ yields

$$U_m \equiv U_m(z=0) = \frac{\pi H}{T}\coth(kh),$$

and substituting this amplitude into the definition of KC gives

$$KC = \frac{U_m T}{D} = \frac{\pi H}{D}\coth(kh).$$

For the operational sea state, with $H_{\rm op} = 1.29\,\mathrm{m}$, $T_{\rm op} = 3.96\,\mathrm{s}$, $h = 93.32\,\mathrm{m}$, and $D = 10\,\mathrm{m}$, the wavenumber $k_{\rm op}$ is obtained from the linear dispersion relation

$$\omega_{\rm op}^2 = g k_{\rm op}\tanh(k_{\rm op}h), \quad \omega_{\rm op} = \frac{2\pi}{T_{\rm op}},$$

which is solved numerically to give $k_{\rm op} \approx 0.257\,\mathrm{m}^{-1}$ and $k_{\rm op}h \approx 24$. Since $k_{\rm op}h \approx 24 > \pi$, the wave lies in deep water according to standard depth classifications, which define deep water as $h/\lambda > 1/2$ (equivalently $kh > \pi$), shallow water as $h/\lambda < 1/20$ (equivalently $kh < \pi/10$), and intermediate depth as $1/20 \le h/\lambda \le 1/2$ [3]. In the deep-water limit, the hyperbolic cotangent tends to unity, $\coth(k_{\rm op}h) \approx 1$, so the near-surface velocity amplitude simplifies to

$$U_{m,\rm op} \approx \frac{\pi H_{\rm op}}{T_{\rm op}} = \frac{\pi \times 1.29}{3.96} \approx 1.0\,\mathrm{m\,s^{-1}}.$$

Substituting this amplitude into the definition of KC gives

$$KC_{\rm op} = \frac{U_{m,\rm op}T_{\rm op}}{D} \approx \frac{1.0 \times 3.96}{10} \approx 0.41.$$

For the extreme storm sea state, with $H_{\rm ex} = 18.62\,\mathrm{m}$, $T_{\rm ex} = 23.79\,\mathrm{s}$, $h = 93.32\,\mathrm{m}$, and $D = 10\,\mathrm{m}$, the dispersion relation

$$\omega_{\rm ex}^2 = g k_{\rm ex}\tanh(k_{\rm ex}h), \quad \omega_{\rm ex} = \frac{2\pi}{T_{\rm ex}},$$

gives $k_{\rm ex} \approx 9.82 \times 10^{-3}\,\mathrm{m}^{-1}$, so that $k_{\rm ex}h \approx 0.92$. This value satisfies $\pi/10 \lesssim k_{\rm ex}h \lesssim \pi$, so the wave is in intermediate depth. In this regime, the hyperbolic cotangent must be retained explicitly, and the near-surface velocity amplitude becomes

$$U_{m,\rm ex} = \frac{\pi H_{\rm ex}}{T_{\rm ex}}\coth(k_{\rm ex}h) \approx \frac{\pi \times 18.62}{23.79} \times 1.38 \approx 3.4\,\mathrm{m\,s^{-1}}.$$

The corresponding Keulegan–Carpenter number is

$$KC_{\rm ex} = \frac{U_{m,\rm ex}T_{\rm ex}}{D} \approx \frac{3.4 \times 23.79}{10} \approx 8.1.$$

Experimental and design-oriented studies of oscillatory flow past cylinders show that the Keulegan–Carpenter number can be used as an indicator of the relative importance of inertia and drag forces in Morison loading. Engineering guidelines typically classify flows with $KC \lesssim 5$ as inertia dominated, flows with $KC \gtrsim 25$–40 as drag dominated, and intermediate values as a mixed regime in which both contributions are important [4]. The operational sea state with $KC_{\rm op} \approx 0.4$ therefore lies firmly in the inertia-dominated regime, while the extreme storm with $KC_{\rm ex} \approx 8$ falls into the mixed regime where drag affects the peak inline force, but inertia still provides the main contribution.

C. Diameter-to-wavelength ratio and Morison applicability

The diameter-to-wavelength ratio D/λ provides a quantitative measure of how slender the monopile is relative to the incident waves and therefore controls whether wave diffraction can be neglected. For a slender cylinder, the hydrodynamic load can be described using the Morison equation, which assumes that the inline force is the sum of a viscous drag term and an inertia term based on the undisturbed wave kinematics at the cylinder centreline. This assumption is only valid if the cylinder does not significantly perturb the surrounding wave field, that is, if diffraction effects remain small.

The wavelength λ associated with a given sea state is obtained from the linear dispersion relation, which has already been solved to provide the wavenumber k for each design condition. For a monochromatic wave in constant depth, the wavelength is

$$\lambda = \frac{2\pi}{k},$$

so that the diameter-to-wavelength ratio can be written explicitly as

$$\frac{D}{\lambda} = \frac{kD}{2\pi}.$$

This expression shows that D/λ is proportional to the non-dimensional diffraction parameter kD, so a small D/λ is equivalent to a small kD.

For the operational sea state, the dispersion calculation in section II-B gives the deep-water wavenumber $k_{\rm op} \approx 0.257\,\mathrm{m}^{-1}$ for the local peak period. Substituting this value into the definition of the wavelength yields

$$\lambda_{\rm op} = \frac{2\pi}{k_{\rm op}} = \frac{2\pi}{0.257} \approx 24.5\,\mathrm{m},$$

so that, with monopile diameter $D = 10\,\mathrm{m}$,

$$\frac{D}{\lambda_{\rm op}} = \frac{10}{24.5} \approx 0.41, \qquad k_{\rm op}D = 2\pi\frac{D}{\lambda_{\rm op}} \approx 2.6.$$

In operation, the monopile diameter is therefore about $41\,\%$ of the incident wavelength and the diffraction parameter $k_{\rm op}D$ is of order 1, which indicates that the cylinder occupies a substantial fraction of the wave crest and will generate a strong diffracted wave.

For the extreme storm sea state, the wavenumber consistent with the longer peak period at the site is $k_{\rm ex} \approx 9.82 \times 10^{-3}\,\mathrm{m}^{-1}$. Using the same linear relation, the corresponding wavelength and diameter-to-wavelength ratio are

$$\lambda_{\rm ex} = \frac{2\pi}{k_{\rm ex}} = \frac{2\pi}{9.82 \times 10^{-3}} \approx 6.40 \times 10^2 m,$$

$$\frac{D}{\lambda_{\rm ex}} = \frac{10}{6.40 \times 10^2} \approx 1.6 \times 10^{-2},$$

$$k_{\rm ex}D = 2\pi\frac{D}{\lambda_{\rm ex}} \approx 0.10.$$

In this extreme storm, the monopile diameter is only about $1.6\,\%$ of the incident wavelength, and the diffraction parameter $k_{\rm ex}D$ is much less than 1, so the structure is very slender relative to the waves.

Comparison with standard slenderness criteria clarifies when the Morison formulation is valid. Linear theory for wave loading on slender cylinders treats the cylinder as slender if its diameter is much smaller than the wavelength, and practical guidance typically recommends that methods based on the Morison equation be used only when

$$\frac{D}{\lambda} \lesssim 0.2 \quad [5].$$

Within this range, the diffracted wave is a small perturbation to the incident field, and the local flow at the cylinder can be approximated as the undisturbed wave kinematics plus a small inertia correction. When D/λ exceeds 0.2, the cylinder blocks a non-negligible portion of the wave crest, diffraction becomes strong, and the Morison equation loses accuracy because its underlying assumption of negligible wave scattering is no longer satisfied.

Applying this criterion to the two design sea states shows that the extreme storm condition, with $D/\lambda_{\mathrm{ex}} \approx 0.016$ and $k_{\mathrm{ex}}D \approx 0.10$, lies well inside the slender-cylinder regime where diffraction is weak and Morison theory is appropriate, provided the Keulegan–Carpenter number confirms inertia-dominated loading [5]. In contrast, the operational sea state, with $D/\lambda_{\mathrm{op}} \approx 0.41$ and $k_{\mathrm{op}}D \approx 2.6$, lies far outside the recommended range: diffraction is expected to dominate the inline force and the Morison equation cannot be used in its original form without large, frequency-dependent corrections.

In this non-slender regime, wave loads are more consistently obtained from linear diffraction theory for a vertical cylinder. The classical MacCamy–Fuchs solution solves the linear potential-flow boundary-value problem for a vertical circular cylinder in regular waves and provides the complex amplitude of the inline force as an explicit function of kD and water depth [6]. Its prediction can be written as an effective inertia coefficient $C_M(kD)$ multiplying the undisturbed horizontal acceleration profile in a Morison-type expression, with an additional viscous drag term added if required. For the operational sea state, the present analysis therefore uses MacCamy–Fuchs to evaluate the inertia-dominated inline force, then represents that result through an effective $C_M(D/\lambda)$ in the Morison formulation, while for the extreme storm, the standard Morison equation with constant coefficients is retained because the structure behaves as a truly slender cylinder.

D. Choice of Morison coefficients

The Morison coefficients for the monopile are chosen using the non-dimensional parameters of section II-B together with laboratory data for oscillatory flow past circular cylinders. The extreme storm case, with $KC_{\mathrm{ex}} \approx 8.1$ and $D/\lambda_{\mathrm{ex}} \ll 1$, satisfies the slenderness criteria for the Morison formulation, and because of the KC number, both inertia and drag terms must be retained. The operational case, with $KC_{\mathrm{op}} \approx 0.41$ but $D/\lambda_{\mathrm{op}} \approx 0.41$, is not slender and therefore uses a diffraction-based MacCamy–Fuchs description for the inertia load, and due to the small value of KC only an effective inertia coefficient is required and the drag term is neglected.

The Reynolds number and its ratio with KC for a vertical cylinder of diameter D in sinusoidal flow with near-surface velocity amplitude U_m and period T are

$$\mathrm{Re} = \frac{U_m D}{\nu}, \qquad \beta = \frac{\mathrm{Re}}{KC} = \frac{D^2}{\nu T},$$

with kinematic viscosity $\nu \approx 1.0\times10^{-6}\,\mathrm{m^2\,s^{-1}}$. For the present monopile, $D = 10\,\mathrm{m}$ gives

$$\beta_{\mathrm{op}} \approx \frac{10^2}{(1.0\times10^{-6})\times 3.96} \approx 2.5\times10^7,$$

$$\beta_{\mathrm{ex}} \approx \frac{10^2}{(1.0\times10^{-6})\times 23.79} \approx 4.2\times10^6.$$

Both sea states therefore lie deep in the high-β regime where hydrodynamic coefficients vary only weakly with further increases in β and are controlled mainly by KC and surface roughness [7], [8].

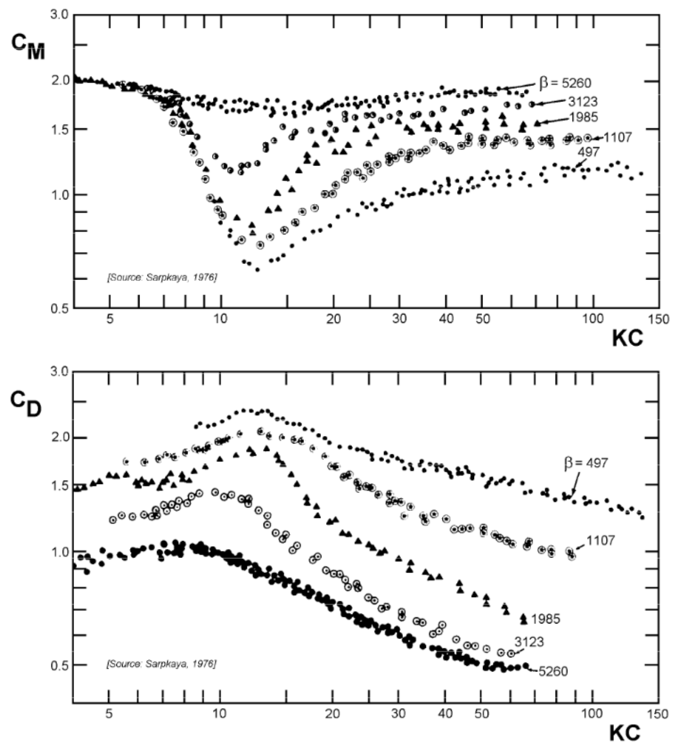

Figure 2: Inertia coefficient C_M (top) and drag coefficient C_D (bottom) as functions of Keulegan–Carpenter number KC for several values of the stability parameter β in oscillatory flow past a circular cylinder [7].

The inertia coefficient is selected first because both sea states lie in the inertia-dominated range $KC \lesssim 10$. Figure 2 shows that for KC between about 0.5 and 10 and $\beta = 5260$ the measured inertia coefficient C_M remains close to 2 [7]. Compilations of oscillatory-flow data and design summaries report the same behaviour and note that for high Reynolds numbers and low to moderate KC the inertia coefficient approaches the potential-flow value $C_M = 2$ for circular cylinders, with recommended design values rarely below about 1.8 [8]. Since β_{op} and β_{ex} exceed the experimental range by several orders of magnitude, it is consistent to treat the laboratory curves as already close to their asymptotic limit and to adopt a single inertia coefficient for both sea states,

$$C_{M,\mathrm{op}} = C_{M,\mathrm{ex}} = 2.0.$$

In the operational case, this C_M is used as an effective inertia coefficient in the MacCamy–Fuchs representation, while in the extreme storm, it is used directly in the Morison inertia term.

The drag coefficient cannot be extrapolated as robustly from figure 2. The measured C_D data at KC of order 1–10 exhibit substantial scatter and strong sensitivity to surface roughness even within the limited range of β covered by the experiments [7], so extending these curves to $\beta \sim 10^6$–10^7 is not reliable. Instead, approximate design values are taken from guidance on hydrodynamic coefficients for offshore tubular members. Design rules and their summaries indicate that for post-critical Reynolds numbers, the drag coefficient for smooth circular cylinders in oscillatory flow should not be taken less than about 0.6–0.7, with values around 0.7 widely used in deterministic global design, and that cylinders with marine growth require drag coefficients at least this large [8]. The present monopile is expected to develop marine roughness over its lifetime, and the extreme load case is governed by inertia with $KC_{\mathrm{ex}} \approx 8.1$, so adopting

$$C_{D,\mathrm{ex}} = 0.7$$

is both consistent with these recommendations and slightly conservative. In the operational sea state, the small value $KC_{\mathrm{op}} \approx 0.41$ implies a negligible drag contribution compared with inertia, so the drag term in the Morison equation is omitted and no specific C_D value is required. Subsequent load calculations therefore use $C_M = 2.0$ in both sea states, retain drag with $C_D = 0.7$ only for the extreme storm, and neglect drag entirely in the operational condition.

E. Overturning moment at seabed level

Wave-induced overturning moments at seabed level in the extreme storm sea state are obtained by integrating the Morison inline force along the submerged monopile. The previous subsections established that the 50-year storm satisfies the slenderness criteria for the Morison formulation and lies in a mixed inertia–drag regime, so both inertia and drag contributions are retained while the cylinder is still treated as dynamically fixed.

The horizontal particle velocity for a regular linear wave of height H, angular frequency ω, wavenumber k, and water depth h is

$$u(z,t) = \frac{H\omega}{2}\,\frac{\cosh\big(k(z+h)\big)}{\sinh(kh)}\cos(kx-\omega t),$$

where z is measured upward from still water level, with $z = 0$ at the free surface and $z = -h$ at the seabed. Differentiating with respect to time gives the horizontal acceleration

$$\frac{\partial u}{\partial t}(z,t) = -\frac{H\omega^2}{2}\,\frac{\cosh\big(k(z+h)\big)}{\sinh(kh)}\sin(kx-\omega t),$$

so the acceleration amplitude is

$$a(z) = \left|\frac{\partial u}{\partial t}\right|_{\max} = \frac{H\omega^2}{2}\,\frac{\cosh\big(k(z+h)\big)}{\sinh(kh)}.$$

Using the dispersion relation $\omega^2 = gk\tanh(kh)$, and $\tanh(kh) = \sinh(kh)/\cosh(kh)$, this can be written as

$$a(z) = \frac{Hgk}{2}\,\frac{\cosh\big(k(z+h)\big)}{\cosh(kh)}.$$

The inline Morison force per unit length on a vertical cylinder of diameter D is

$$f(z,t) = \rho C_M \frac{\pi D^2}{4}\,\frac{\partial u}{\partial t}(z,t) + \tfrac{1}{2}\,\rho C_D D\,\big|u(z,t)\big|u(z,t),$$

so the maximum inertia force per unit length is

$$\begin{aligned} f_{I,\max}(z) &= \rho C_M \frac{\pi D^2}{4}\,a(z) \\ &= \rho C_M \frac{\pi D^2}{4}\,\frac{Hgk}{2}\,\frac{\cosh\big(k(z+h)\big)}{\cosh(kh)}. \end{aligned}$$

The corresponding overturning moment at the seabed is obtained by integrating this distributed force multiplied by its lever arm $(z+h)$ with respect to the seabed at $z = -h$,

$$M_I = \int_{-h}^{0} f_{I,\max}(z)\,(z+h)\,\mathrm{d}z.$$

Introducing the change of variable $y = z + h$ (so $z = y - h$ and $y \in [0,h]$) yields

$$M_I = \rho C_M \frac{\pi D^2}{4}\,\frac{Hgk}{2}\,\frac{1}{\cosh(kh)}\int_0^h y\cosh(ky)\,\mathrm{d}y.$$

The remaining integral is evaluated by parts:

$$\begin{aligned} \int_0^h y\cosh(ky)\,\mathrm{d}y &= \Big[\frac{y}{k}\sinh(ky)\Big]_0^h - \int_0^h \frac{1}{k}\sinh(ky)\,\mathrm{d}y \\ &= \frac{h}{k}\sinh(kh) - \frac{1}{k^2}\big[\cosh(ky)\big]_0^h \\ &= \frac{1}{k^2}\Big[kh\sinh(kh) - \cosh(kh) + 1\Big]. \end{aligned}$$

Substituting this expression gives the closed-form inertia-induced overturning moment

$$M_I = \rho C_M \frac{\pi D^2 H g}{8k\cosh(kh)}\Big[kh\sinh(kh) - \cosh(kh) + 1\Big].$$

The drag term is evaluated using the velocity amplitude $u_m(z)$, defined by

$$\begin{aligned} u(z,t) &= u_m(z)\cos(kx-\omega t), \\ u_m(z) &= \frac{H\omega}{2}\,\frac{\cosh\big(k(z+h)\big)}{\sinh(kh)}. \end{aligned}$$

The maximum drag force per unit length occurs when $\cos(kx-\omega t) = \pm 1$ and has magnitude

$$\begin{aligned} f_{D,\max}(z) &= \tfrac{1}{2}\,\rho C_D D\,u_m^2(z) \\ &= \tfrac{1}{2}\,\rho C_D D\,\Big[\frac{H\omega}{2}\Big]^2 \frac{\cosh^2\big(k(z+h)\big)}{\sinh^2(kh)}. \end{aligned}$$

The corresponding overturning moment is

$$\begin{aligned} M_D &= \int_{-h}^{0} f_{D,\max}(z)\,(z+h)\,\mathrm{d}z \\ &= \tfrac{1}{2}\,\rho C_D D\Big[\frac{H\omega}{2}\Big]^2 \frac{1}{\sinh^2(kh)} \\ &\quad \int_{-h}^{0} (z+h)\cosh^2\big(k(z+h)\big)\,\mathrm{d}z. \end{aligned}$$

With the change of variable $y = z + h$, this becomes

$$M_D = \tfrac{1}{2}\,\rho C_D D \Big[\frac{H\omega}{2}\Big]^2 \frac{1}{\sinh^2(kh)} \int_0^h y\cosh^2(ky)\,\mathrm{d}y.$$

Using $\cosh^2(ky) = \frac{1}{2}\big[1 + \cosh(2ky)\big]$ and integrating term by term yields

$$\begin{aligned}\int_0^h y\cosh^2(ky)\,\mathrm{d}y &= \frac{1}{2}\int_0^h y\,\mathrm{d}y + \frac{1}{2}\int_0^h y\cosh(2ky)\,\mathrm{d}y \\ &= \frac{h^2}{4} + \frac{1}{2}\Big[\frac{y\sinh(2ky)}{2k} - \frac{\cosh(2ky)}{4k^2}\Big]_0^h \\ &= \frac{H^2\omega^2}{16k^2}\Big[h^2k^2\,\mathrm{csch}^2(kh) + 2hk\,\coth(kh) - 1\Big],\end{aligned}$$

which simplifies the drag-induced overturning moment to

$$M_D = \rho C_D D\,\frac{H^2\omega^2}{32k^2}\Big[h^2k^2\,\mathrm{csch}^2(kh) + 2hk\,\coth(kh) - 1\Big],$$

where $\mathrm{csch}(kh) = 1/\sinh(kh)$ and $\coth(kh) = \cosh(kh)/\sinh(kh)$.

The extreme seabed moment is taken as the sum of the inertia and drag contributions,

$$M_{\text{wave,ex}} = M_{I,\text{ex}} + M_{D,\text{ex}},$$

which assumes that the peaks of the inertia and drag components act in phase. This neglects the nominal 90° phase shift between inertia and drag in regular waves but provides a conservative estimate appropriate for ultimate-limit-state design.

For the extreme storm sea state in table II, the inputs are $H_{\text{ex}} = 18.62\,\mathrm{m}$, $T_{\text{ex}} = 23.79\,\mathrm{s}$, $h = 93.32\,\mathrm{m}$, $D = 10\,\mathrm{m}$, $\rho = 1025\,\mathrm{kg\,m^{-3}}$, $g = 9.81\,\mathrm{m\,s^{-2}}$, $C_M = 2.0$, and $C_D = 0.7$. The dispersion relation

$$\omega_{\text{ex}}^2 = g k_{\text{ex}} \tanh(k_{\text{ex}} h), \qquad \omega_{\text{ex}} = \frac{2\pi}{T_{\text{ex}}},$$

gives $k_{\text{ex}} \approx 9.82 \times 10^{-3}\,\mathrm{m^{-1}}$ and $k_{\text{ex}}h \approx 0.92$, corresponding to intermediate depth. Substituting these values into the expressions above yields an inertia-induced overturning moment

$$M_{I,\text{ex}} \approx 5.3 \times 10^8\,\mathrm{N\,m},$$

and a drag-induced overturning moment

$$M_{D,\text{ex}} \approx 1.3 \times 10^8\,\mathrm{N\,m}.$$

The total wave-induced seabed moment in the 50-year storm is therefore

$$M_{\text{wave,ex}} \approx M_{I,\text{ex}} + M_{D,\text{ex}} \approx 6.6 \times 10^8\,\mathrm{N\,m},$$

with drag contributing about one quarter of the inertia component. This confirms that inertia dominates the extreme storm seabed bending while drag provides a non-negligible correction that must be included for a consistent Morison-based design in the mixed Keulegan–Carpenter regime.

In the operational sea state, a direct Morison-based evaluation of the seabed moment is not appropriate because the monopile is not slender relative to the local wavelength. The diameter-to-wavelength analysis above gave $D/\lambda_{\text{op}} \approx 0.41$ and $k_{\text{op}}D \approx 2.6$, which exceed the usual slender-cylinder threshold $D/\lambda \lesssim 0.2$ for which diffraction effects remain small and the pure Morison equation remains accurate. For this geometry, the monopile blocks a substantial portion of the wave crest and generates a strong scattered field, so the inline load is more consistently obtained from a diffraction solution and only then represented in Morison form.

To retain the closed-form expression for the seabed moment while accounting for diffraction, the MacCamy–Fuchs solution is used to define a frequency-dependent effective inertia coefficient $C_M^{\text{MF}}(kD)$. The coefficient for a vertical surface-piercing circular cylinder is written as

$$C_M^{\text{MF}}(kD) = \frac{4}{\pi (kr)^2 \sqrt{J_{01}^2(kr) + Y_{01}^2(kr)}},$$

where $r = D/2$ is the cylinder radius and

$$J_{01}(x) = J_0(x) - \frac{1}{x}J_1(x), \qquad Y_{01}(x) = Y_0(x) - \frac{1}{x}Y_1(x),$$

with J_n and Y_n the Bessel functions of the first and second kind of order n [9]. In this representation, the inline inertia force per unit length retains the Morison form $\rho\, C_M^{\text{MF}} \pi D^2 a(z)/4$, but $C_M^{\text{MF}}(kD)$ captures the diffraction-induced modification of the added-mass force relative to the slender limit.

Evaluating this expression for the Bretagne Sud 1 sea states links the diffraction description back to the Morison inertia term. For the operational case, the dispersion relation yields $k_{\text{op}} \approx 0.257\,\mathrm{m^{-1}}$, so that $k_{\text{op}}D \approx 2.57$ and

$$C_M^{\text{MF}}(k_{\text{op}}D) \approx 1.03.$$

For the extreme storm, $k_{\text{ex}} \approx 9.82 \times 10^{-3}\,\mathrm{m^{-1}}$ gives $k_{\text{ex}}D \approx 0.10$ and

$$C_M^{\text{MF}}(k_{\text{ex}}D) \approx 2.01,$$

essentially identical to the potential-flow value $C_M \approx 2$ associated with the Froude–Krylov plus added-mass limit. The earlier choice $C_M = 2.0$ for the storm case is therefore consistent with MacCamy–Fuchs, and an effective inertia coefficient $C_{M,\text{op}} = 1.03$ is adopted for the operational case.

With drag neglected in operation because $KC_{\text{op}} \approx 0.41$, the operational overturning moment is obtained by inserting the operational parameters into the inertia-only closed-form expression for the seabed moment derived above. Using $H_{\text{op}} = 1.29\,\mathrm{m}$, $T_{\text{op}} = 3.96\,\mathrm{s}$, $h = 93.32\,\mathrm{m}$, $D = 10\,\mathrm{m}$, $\rho = 1025\,\mathrm{kg\,m^{-3}}$, $g = 9.81\,\mathrm{m\,s^{-2}}$, $k_{\text{op}} \approx 0.257\,\mathrm{m^{-1}}$ from the dispersion relation, and the effective inertia coefficient $C_{M,\text{op}} = 1.03$ gives an inertia-induced operational seabed moment

$$M_{I,\text{op}} \approx 4.7 \times 10^7\,\mathrm{N\,m},$$

with the vertical distribution of the load governed by the same Airy-wave acceleration profile as in the storm calculation.

Comparison with the extreme storm moment shows that operational waves do not govern the monopile design. The 50-year storm analysis above gave a total wave-induced seabed moment $M_{\text{wave,ex}} \approx 6.6 \times 10^8\,\mathrm{N\,m}$ when inertia and drag are summed in phase, so the operational inertia moment is smaller by a factor of about $M_{\text{wave,ex}}/M_{I,\text{op}} \approx 14$. Even when only the storm inertia term is considered, $M_{I,\text{ex}} \approx 5.3 \times 10^8\,\mathrm{N\,m}$ still exceeds $M_{I,\text{op}}$ by a factor of about 11. This disparity reflects both the ratio of wave heights $H_{\text{op}}/H_{\text{ex}} \approx 0.07$ and the reduction in C_M^{MF} at $kD \approx 2.6$, and indicates that in normal operation global bending is controlled by aerodynamic thrust and associated hydrodynamic response rather than by operational wave loading alone.

F. Overturning moment due to wind

Wind-induced overturning moments are now compared with wave-induced moments to identify the dominant loading mechanism at the seabed level in each sea state. The operational and extreme wind speeds are those in table II, obtained from long-term statistics at sea level and extrapolated to the hub height of 120 m using the power-law profile in equation (1).

The vertical lever arm for wind loads is set by the distance between the monopile toe and the hub. With the seabed at $z = -h = -93.32\,\mathrm{m}$ and the hub at $z_{\mathrm{hub}} = 120\,\mathrm{m}$ above mean sea level, the moment arm for a horizontal hub force is

$$L = h + z_{\mathrm{hub}} = 93.32 + 120 \approx 213.3\,\mathrm{m}.$$

This lever arm is used for both the operational and extreme wind cases, because the hub height and water depth are unchanged between sea states.

In operation, the turbine is assumed to run near rated conditions with blades pitched for power production, so wind loading at the seabed is dominated by the rotor thrust. The thrust is modelled as a drag force acting on the rotor swept area $A_{\mathrm{rot}} = \pi D_{\mathrm{rot}}^2/4$, with an effective thrust coefficient C_D chosen to represent near-rated aerodynamic loading,

$$T_{\mathrm{op}} = \tfrac{1}{2}\,\rho_{\mathrm{air}}\,C_D\,A_{\mathrm{rot}}\,U_{\mathrm{hub,op}}^2,$$

where $\rho_{\mathrm{air}} \approx 1.225\,\mathrm{kg\,m^{-3}}$, $D_{\mathrm{rot}} \approx 178.3\,\mathrm{m}$ for the DTU 10 MW turbine [2], $U_{\mathrm{hub,op}} = 9.49\,\mathrm{m\,s^{-1}}$, and an effective coefficient $C_D = 0.75$ is adopted to be consistent with a thrust close to the rated value. Substituting these values gives a rotor area $A_{\mathrm{rot}} \approx 2.5 \times 10^4\,\mathrm{m}^2$ and an operational thrust

$$T_{\mathrm{op}} \approx 1.0 \times 10^6\,\mathrm{N},$$

which is of the same order as the maximum thrust $T_{\max} = 1.5 \times 10^6$ N from table I and therefore consistent with near-rated operation.

The corresponding operational wind-induced overturning moment at the seabed follows as

$$M_{\mathrm{wind,op}} = T_{\mathrm{op}}\,L \approx 1.0 \times 10^6 \times 2.13 \times 10^2 \approx 2.2 \times 10^8\,\mathrm{N\,m}.$$

The wave-induced seabed moment for the operational sea state was obtained in section II by applying the MacCamy–Fuchs effective inertia coefficient with drag neglected because $KC_{\mathrm{op}} \approx 0.41$, giving

$$M_{\mathrm{wave,op}} = M_{I,\mathrm{op}} \approx 4.7 \times 10^7\,\mathrm{N\,m}.$$

The ratio of wind to wave moments in operation is therefore

$$\frac{M_{\mathrm{wind,op}}}{M_{\mathrm{wave,op}}} \approx \frac{2.2 \times 10^8}{4.7 \times 10^7} \approx 4.7,$$

so aerodynamic thrust produces an overturning moment almost five times larger than the operational wave-induced moment. In normal operation, seabed bending is thus controlled by wind loading on the rotor and tower, with operational waves contributing only about 20 % of the combined wind–wave moment.

In the extreme storm sea state, the turbine is assumed parked and feathered so that the rotor drag is minimised. The parked wind load is represented using the parked drag coefficient $C_{D,\mathrm{turb}} = 0.01$ from table I,

$$T_{\mathrm{park}} = \tfrac{1}{2}\,\rho_{\mathrm{air}}\,C_{D,\mathrm{turb}}\,A_{\mathrm{rot}}\,U_{\mathrm{hub,ex}}^2,$$

where $U_{\mathrm{hub,ex}} = 39.2\,\mathrm{m\,s^{-1}}$ is the 50-year hub-height wind speed. Using the same ρ_{air} and A_{rot} as above gives

$$T_{\mathrm{park}} \approx 2.3 \times 10^5\,\mathrm{N},$$

and hence a parked-wind overturning moment at the seabed

$$M_{\mathrm{wind,ex}} = T_{\mathrm{park}}\,L \approx 5.0 \times 10^7\,\mathrm{N\,m}.$$

The corresponding wave-induced seabed moment in the 50-year storm, combining inertia and drag contributions in phase, was found in section II as

$$\begin{aligned} M_{\mathrm{wave,ex}} &\approx M_{I,\mathrm{ex}} + M_{D,\mathrm{ex}} \\ &\approx 5.3 \times 10^8 + 1.3 \times 10^8 \\ &\approx 6.6 \times 10^8\,\mathrm{N\,m}. \end{aligned}$$

The wind–wave moment ratio in the storm case is therefore

$$\frac{M_{\mathrm{wind,ex}}}{M_{\mathrm{wave,ex}}} \approx \frac{5.0 \times 10^7}{6.6 \times 10^8} \approx 0.08,$$

so storm waves generate an overturning moment at the seabed more than an order of magnitude larger than the parked-wind load.

Taken together, these comparisons show that different mechanisms dominate in the two design conditions. In operation, the seabed moment is governed by aerodynamic thrust, with waves providing a secondary contribution that may still matter for fatigue but not for ultimate strength. In the 50-year storm, the monopile experiences a wave-driven ultimate loading scenario in which Morison-type hydrodynamic forces dominate and parked-wind effects can be treated as a small superimposed contribution.

III. Waveloads on a Floating Offshore Wind Turbine

A. Input Parameters

In addition to the design environmental conditions described in subsection II-A (H and T), the 50-year return period windspeed has been assessed using a Block Maxima (BM) method applied to the 27-year Resourcecode dataset described in Parts 1 and 2 of this project. To determine the corresponding wind speed at the turbine hub height of 120 m, the reference wind speed at 10 m was extrapolated using the power law wind profile, as was shown in equation 1.

A shear exponent of $\alpha = 0.11$ was selected for this extrapolation. This value is widely recognised for open water conditions under neutral atmospheric stability, as recommended by standards such as IEC61400-3 [10]. This coefficient reflects the lower surface roughness characteristic of the offshore marine boundary layer compared to onshore environments.

In addition to the wind profile, the aerodynamic loads were modelled using state-dependent drag coefficients to reflect the turbine's operational status accurately. For standard operational conditions, a drag coefficient of $C_D = 0.75$ was applied to represent the rotor thrust. For extreme storm conditions, defined by wind speeds exceeding the turbine's cut-out threshold ($U >$ 25 m/s), the turbine is considered parked with blades feathered. In this survival mode, a reduced drag coefficient of $C_D = 0.01$ was implemented, in accordance with the turbine specifications defined in the project documentation.

B. Results and Interpretation

Feasibility assessments were subsequently conducted for both operational and extreme scenarios. Simulations under mean wind and wave conditions generally demonstrated stable dynamic responses, with surge drift stabilised by the mooring system. However, the analysis under 50-year extreme storm conditions ($H_S = 18.62$ m, $T_p = 23.79$ s) revealed distinct critical instabilities for each floater concept, driven by their respective natural frequencies.

1) SPAR Platform Analysis: The time-domain response of the SPAR platform under extreme conditions is presented in Figure 3. The results reveal a critical instability in the heave degree of freedom.

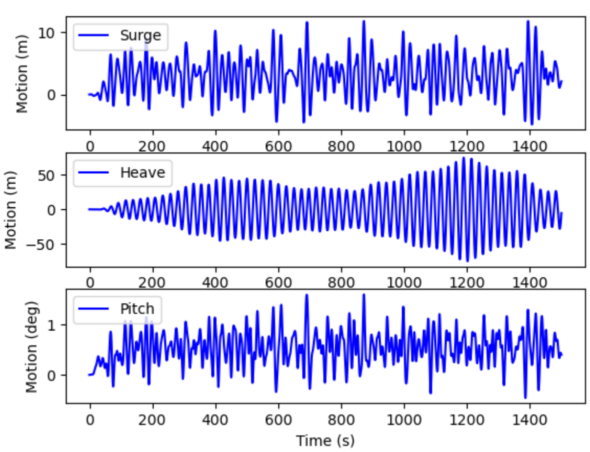

Figure 3: Time-domain response of the SPAR floater under 50-year extreme storm conditions. Note the pronounced amplification in heave.

The heave motion exhibits a pronounced amplification phenomenon, where amplitudes grow excessively to values of ± 60 m. This failure is attributed to resonance caused by the coincidence of the wave excitation period ($T_p = 23.79$ s) with the platform's natural heave period. Verification of this resonance was performed by calculating the natural heave period (T_n) using the system's total mass ($M + A_{33}$) and total vertical stiffness (K_{33}) derived from the hydrodynamic dataset:

$$T_{n,heave}^{SPAR} = 2\pi\sqrt{\frac{M + M_A}{K_H}} \approx 24.4 \text{ s} \tag{2}$$

Substituting the values from the model ($M + A_{33} \approx 1.24 \times 10^7$ kg, $K_{33} \approx 8.21 \times 10^5$ N/m) yields $T_{n,heave}^{SPAR} \approx 24.4$ s. The proximity of this natural period to the peak spectral period of the storm confirms that the initial spar geometry is resonating with the storm energy. Conversely, the pitch motion remains stable and non-resonant, as the natural pitch period ($T_n \approx 43$ s) is well separated from the wave energy spectrum.

2) BARGE Platform Analysis: The time-domain response of the BARGE platform is presented in Figure 4. In contrast to the SPAR, the Barge exhibits a critical response in the pitch degree of freedom.

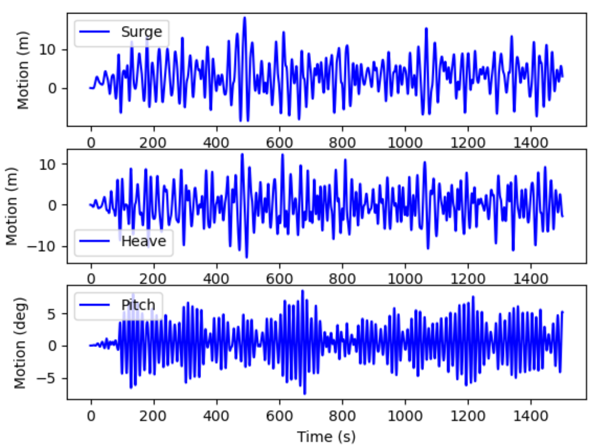

Figure 4: Time-domain response of the BARGE floater. Note the resonant behaviour in Pitch.

The pitch response displays resonance characteristics with modulated amplitude envelopes. This behaviour may be explained by the platform's natural pitch period:

$$T_{n,pitch}^{BARGE} = 2\pi\sqrt{\frac{I_{55} + A_{55}}{K_{55}}} \approx 15.2 \text{ s} \tag{3}$$

Although the storm's peak period ($T_p = 23.79$ s) is distinct from this natural period, the irregular JONSWAP spectrum distributes wave energy across a broad frequency band. The spectrum might retain sufficient energy at $T = 15$ s to excite the Barge's pitch mode. Unlike the SPAR, which resonates due to the spectral peak, the Barge is excited by the spectral tail. This makes the Barge inherently susceptible to resonance not only in this extreme case but also in common sea states where the peak period falls between 12–16 s.

3) Summary of Observations: Both concepts demonstrate stable low-frequency surge motions ($T_n > 100$ s) driven by the mooring system. However, the specific geometry of each floater induces different failure modes: the SPAR fails due to direct resonance with the 50-year peak period, while the BARGE suffers from pitch resonance excited by the broadband energy of the wave spectrum, exceeding the maximum acceptable pitch of $5°$.

IV. CONCLUSION

This report linked the Bretagne Sud 1 sea states derived from the site-grid ResourceCode hindcast to structural demand

and platform operability through two complementary analyses. An idealised monopile benchmark was used to quantify seabed shear and overturning moments under operational and 50-year storm conditions, while time-domain simulations were used to evaluate the surge, heave, and pitch responses of SPAR and BARGE floating concepts under the same environmental forcing.

The results show a clear regime shift in the dominant loading mechanism. For the monopile exercise, aerodynamic thrust governs the operational seabed overturning demand, whereas storm-wave hydrodynamics dominate the ultimate-limit scenario by more than an order of magnitude relative to parked-wind loading. Although a fixed monopile is not a realistic foundation choice at $h = 93.32\,\mathrm{m}$, the calculation provides a useful reference level for wave-driven bending and a consistent separation of wind- and wave-driven contributions.

The floating-platform analysis highlights that feasibility at Bretagne Sud 1 is constrained primarily by resonance with wave energy rather than by mean drift alone. The SPAR exhibited a heave instability because its natural heave period lay close to the storm peak period, while the BARGE exceeded acceptable pitch limits due to excitation of its pitch natural period by spectral energy away from the peak. These outcomes imply that concept selection and early sizing should prioritise detuning of natural periods from energetic sea-state bands and increasing effective damping or restoring characteristics, for example, via geometry and mass distribution adjustments, hydrostatic stiffness changes, and mooring-system retuning. Further work should extend the present approach to fully coupled aero-hydro-servo-elastic simulations with irregular directional seas, nonlinear wave effects, and fatigue-relevant load metrics to confirm survivability and operational robustness under the full range of site conditions.

Declaration of AI Use

ChatGPT (OpenAI) was used exclusively to refine the manuscript's language, formatting, and structure. All scientific interpretations, analyses, and conclusions remain the sole responsibility of the author.

Acknowledgements

We thank Prof. Marissa Yates, Prof. Jeffrey Harris, and Prof. Christophe Peyrard for doing an excellent job teaching the *Sea States, Wave Propagation, and Ocean Wave Energy* course. We are particularly thankful for their efforts in thoroughly reading through our reports and providing thoughtful feedback. Code and notebooks were jointly maintained at https://github.com/Lucaschikashy/sea-states_project_bretagne.git.

References

[1] C. Peyrard, "TP2_Morison: Overturning moment applied on a monopile supporting the academic DTU 10 MW wind turbine." Laboratory handout, ENSTA Paris, 2025.

[2] C. Bak, F. Zahle, R. Bitsche, T. Kim, A. Yde, L. C. Henriksen, M. H. Hansen, and A. Natarajan, "Description of the dtu 10 mw reference wind turbine," Tech. Rep. Report-I-0092, DTU Wind Energy, July 2013.

[3] J. Harris, C. Peyrard, and M. Yates, "Sea states, wave propagation, and ocean wave energy: Class notes, 2025–2026." Course notes for MEC53456EP *Sea States, Wave Propagation and Ocean Wave Energy*, LHSV, École nationale des ponts et chaussées and EDF R&D, 2025. Document: SSWPOWE_Class_Notes.pdf.

[4] Z. Gao, M. Efthymiou, L. Cheng, T. Zhou, M. Minguez, and W. Zhao, "Hydrodynamic damping of a circular cylinder at low kc: Experiments and an associated model," *Marine Structures*, vol. 72, p. 102777, 2020.

[5] T. Sarpkaya, "Wave forces on offshore structures," *Wave Forces on Offshore Structures*, pp. 1–322, 01 2012.

[6] R. C. MacCamy and R. A. Fuchs, "Wave forces on piles: A diffraction theory," Technical Memorandum 69, U.S. Army Corps of Engineers, Beach Erosion Board, 1954.

[7] T. Sarpkaya, "Force on a circular cylinder in viscous oscillatory flow at low keulegan—carpenter numbers," *Journal of Fluid Mechanics*, vol. 165, p. 61–71, 1986.

[8] B. M. Sumer and J. Fredsøe, *Hydrodynamics Around Cylindrical Structures*. Singapore: World Scientific, 2006.

[9] DNV GL AS, "Environmental conditions and environmental loads," Recommended Practice DNVGL-RP-C205, DNV GL AS, Høvik, Norway, Aug. 2017.

[10] International Electrotechnical Commission, "Wind energy generation systems – part 3-1: Design requirements for fixed offshore wind turbines," Tech. Rep. IEC 61400-3-1:2019, International Electrotechnical Commission (IEC), Geneva, Switzerland, 2019. Publication date: 2019-04-05.